Monomial Matrix Relocation on the Loss Function Level-Set of Feedforward Neural Networks

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***Abstract*—Gradient-based optimizers are crucial for the success of deep learning methods; however, such optimizers frequently suffer from slow convergence and can become trapped in suboptimal local minima or plateaus. To overcome this issue, we propose a novel optimization technique called Monomial Equivalence - Interleaved Training (ME-IT) to augment well-known optimizers. ME-IT periodically applies a monomial matrix transform—a computationally cheap, loss-preserving transformation that leverages the inherent symmetries of the network. This "re-locates" the model to a functionally identical but parametrically different location on the loss level-set, enabling the network to explore the loss surface and find favorable points for continued gradient descent. Experiments on the MNIST dataset demonstrate that ME-IT-applied models converge significantly faster and discover final solutions with lower loss and higher accuracy than their state-of-the-art counterparts. A test accuracy improvement of $\sim 0.3\%$ is achieved on the MNIST dataset with respect to the model with Cosine Annealing with Warm Restarts, and a training speedup of around $2\times$ is reached for shorter training sessions. This work establishes that strategically leveraging network symmetries is a practical tool for improving and accelerating deep learning optimization.**

I. Introduction

Successful Deep Learning (DL) methods rely on the efficacy of gradient-based optimizers. Even though optimizers like Stochastic Gradient Descent (SGD) or Adam [1] have been deployed to a great extent for training complex neural architectures, descending a non-convex landscape with such methods

eter updates via computationally inexpensive, loss-preserving monomial transforms. This allows the network to continue its descend from a different point on the same loss level set. Thus, our contributions in this work are threefold:

1) We propose *ME-IT* algorithm, a novel method that interleaves transforms between monomial equivalent networks and parameter updates to effectively escape loss plateaus and suboptimal parameter configurations.
2) We empirically demonstrate that training standard models with ME-IT leads to significantly faster convergence rates and the discovery of better final solutions on benchmark classification tasks using the MNIST [4] dataset.
3) We show that the resulting models achieve not only lower final training loss but also higher test accuracy, indicating that the ability to escape poor local minima translates to improved generalization.

In Section II, we present a review of existing network symmetries and studies on loss surfaces. In Section III, we describe ME-IT algorithm with key observations. Then, we compare ME-IT with *Cosine Annealing with Warm Restarts* (CAWR) learning rate scheduler empirically and conduct ablation studies on ME-IT parameters in Section IV.